

Multisite Veno-Venous Cannulation for Neonates and Nonambulatory Children*

OBJECTIVES: Neonates with respiratory failure are ideally supported with veno-venous rather than veno-arterial extracorporeal membrane oxygenation due to the reduced rate of neurologic complications. However, the proportion of neonates supported with veno-venous extracorporeal membrane oxygenation is declining. We report multisite veno-venous extracorporeal membrane oxygenation, accessing the neck, returning to the inferior vena cava via the common femoral vein in neonates and children less than 10 kg.

DESIGN: Retrospective case series with 1 year minimum follow-up.

PATIENTS: Patients less than 10 kg supported with veno-venous extracorporeal membrane oxygenation accessing the jugular and returning to the femoral vein

SETTING: A 30-bed pediatric intensive care delivering extracorporeal membrane oxygenation to approximately 20 children annually.

INTERVENTIONS: Veno-venous extracorporeal membrane oxygenation accessing the jugular and returning to the femoral vein was delivered using two single lumen cannulae.

MEASUREMENTS AND MAIN RESULTS: January 2015 to August 2019, 11 patients underwent veno-venous extracorporeal membrane oxygenation accessing the jugular and returning to the femoral vein with median weight of 3.6 kg (interquartile range 2.8–6.1 kg), and median corrected gestational age of 13 days (interquartile range, 2–175 d). The smallest patient weighed 2.1 kg. Seven patients had comorbidities. Extracorporeal membrane oxygenation was technically successful in all patients with median flows of 126 mL/kg/min (interquartile range, 120–138 mL/kg/min) and median arterial oxygenation saturation of 94% (interquartile range, 91–98%) at 24 hours. Nine survived to home discharge, and two were palliated. Common femoral vein occlusion was observed in all patients on ultrasound post decannulation. There was no clinical or functional deficit in the cannulated limb at follow-up, a minimum of 1 year post extracorporeal membrane oxygenation.

CONCLUSIONS: Veno-venous extracorporeal membrane oxygenation accessing the jugular and returning to the femoral vein was performed safely in patients under 10 kg with the smallest patient weighing 2.1 kg. Although occlusion of the common femoral vein was observed in patients post decannulation, subsequent follow-up demonstrated no clinical implications. We challenge current practice that veno-venous extracorporeal membrane oxygenation accessing the jugular and returning to the femoral vein cannot be performed in nonambulatory patients and suggest that this strategy is preferred over veno-arterial extracorporeal membrane oxygenation in infants requiring extracorporeal membrane oxygenation for respiratory failure.

KEY WORDS: extracorporeal membrane oxygenation; multisite; neonate; venovenous

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AT THE BEDSIDE

- V_j-V_f was performed safely in patients under 10 kg with the smallest patient weighing only 2.1 kg. Although occlusion of the common femoral vein was observed in patients post decannulation, subsequent follow-up demonstrated no clinical implications.
- This is the first case series describing V_j-V_f in nonambulatory patients with comprehensive follow-up and outcome data.
- V_j-V_f should be preferred over V-A ECMO in infants requiring ECMO for respiratory failure. Further evaluation of this technique is required with regard to comparison with V-V dual-lumen ECMO.

Neonates and children with refractory respiratory failure can be supported with veno-venous (V-V) or veno-arterial (V-A) extracorporeal membrane oxygenation (ECMO). Neonatal V-A ECMO is commonly provided through cannulating the right internal jugular vein (RIJ) to remove deoxygenated blood from the right atrium (RA) and return oxygenated blood via the carotid artery. V-A ECMO has significantly higher rates of neurologic complications than V-V ECMO, particularly in neonates (1). V-V ECMO can be provided using one dual-lumen cannula (VVDL) at a single site (usually right RIJ) or via multiple sites with single lumen cannulae. V-V ECMO accessing the jugular and returning to the femoral vein (V_j-V_f) has not been advised in children less than 10–15 kg due to concerns that femoral veins (FVs) are too small to cannulate until children are ambulatory (2, 3).

V-V ECMO is the standard of care in adults with respiratory failure, with over 90% supported in this way. Use of V-V ECMO has declined significantly in the neonatal population over the last decade with only 16% of neonates with respiratory failure supported by this mode in 2019 (4). VVDL have been used widely to provide V-V ECMO in neonates and children, but there is currently only one brand of neonatal VVDL cannula

available, Avalon (Getinge, Sweden). VVDL are wire-reinforced and have been associated with cardiac perforation in neonates (5, 6). The company supplying the other previously widely used brand, Origen (Austin, TX), recalled all cannulae in 2018, with no schedule for their product to return to market (7). A recent survey of U.K. ECMO centers cited concerns over cannulae suitability for VVDL and risk of cardiac/vessel perforation as the main reasons for opting for V-A over V-V ECMO for neonates with respiratory failure (8). In this survey, no other center considered V_j-V_f to be feasible in patients less than 10 kg due to concerns that FVs in nonambulatory children are too small. The ELSO “Red Book,” the main ECMO reference manual, states that adequate ECMO flows cannot be achieved with this approach. The risk of damage to the leg due to neurovascular compromise has also been cited. There is a paucity of literature describing the safety of V_j-V_f in smaller children, with a 1980s case series concluding that V-A ECMO was preferential to V_j-V_f in neonates (9).

This article describes the experience of a single centre in using V_j-V_f in children less than 10 kg, delivering V-V ECMO to children down to 2.1 kg (Fig. 1).



RESEARCH IN CONTEXT

- Neonates with respiratory failure are increasingly supported with veno-arterial extracorporeal membrane oxygenation (ECMO) due to concerns regarding the suitability of dual-lumen cannulae. Worldwide, use of veno-venous (V-V) ECMO in neonates with respiratory failure has declined to 16%.
- The alternative is to deliver ECMO using V-V ECMO accessing the jugular and returning to the femoral vein (V_j-V_f), but this has not been advised in infants due to concerns that the femoral vein is too small to provide adequate ECMO flow and avoid limb injury.
- This case series describes 11 patients under 10 kg using V_j-V_f , demonstrating feasibility and no long-term morbidity due to this mode of ECMO support.

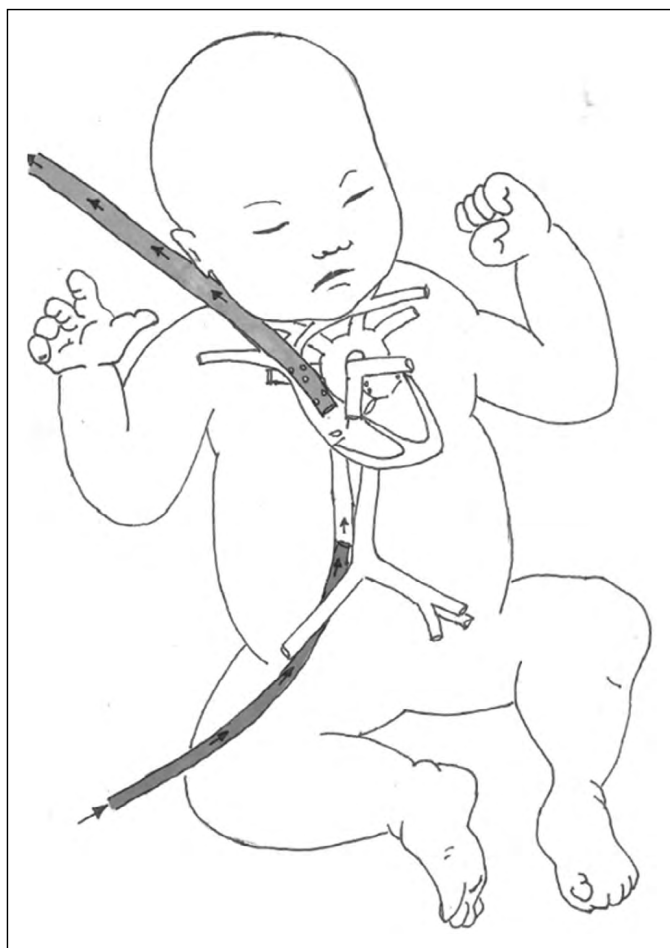


Figure 1. Diagram of extracorporeal membrane oxygenation (ECMO) cannulation configuration used by Evelina London to deliver veno-venous ECMO to neonates and children less than 10 kg.

METHOD

Evelina London Children's Hospital (ELCH) PICU is the second largest PICU in the United Kingdom with approximately 1,200 admissions and 20 ECMO runs annually (75–90th centile in Europe by case numbers [4]). In 2015, the hospital's Treatment, Risk and Assessment Committee agreed that it was preferable to support all children with respiratory failure with V-V ECMO rather than V-A ECMO. Children presenting to ELCH with respiratory failure requiring ECMO were assessed for V_j - V_f cannulation using ultrasound to ensure FV patency. Cannulation was performed surgically by one of three pediatric cardiothoracic surgeons.

Surgical Procedure

The RIJ is accessed, and blood returned to the common FV (CFV) as the majority of cardiac output returns to the heart via the superior vena cava (SVC) rather than

inferior vena cava (IVC) at this age; this ratio can be as high as 4:1 in neonates. The RIJ is surgically exposed, heparin is administered (75 U/kg), and the cannula advanced into the RA. The tip of the access cannula is ideally placed at the level of the SVC/RA junction under ECHO guidance.

The return cannulation site is preferably via the right CFV, having a shorter and straighter course to the IVC than the left. The femoral artery is palpated, and a 1 cm transverse incision made just below the inguinal ligament. The CFV is identified, controlled, and a longitudinal purse-string made. An 8F cannula can be inserted into a neonatal CFV. With a semi-Seldinger technique, the needle is introduced into the skin 3–4 cm below the surgical incision and tunnelled into the CFV. The dilator and cannula are inserted over the guide wire and advanced into the abdominal IVC, to reach the 5–6 cm mark at the venous entry point (**Fig. 2A**). The semi-Seldinger technique has also been used for RIJ cannulation. Preexisting central catheters at cannulation sites were resited, not rewired.

Equipment used are as follows: cannulae (BioMedicus; Medtronic Inc, Minneapolis, MN), centrifugal pump (Centrimag; Thoratec Switzerland GmbH, Zurich, Switzerland), Bioline (heparin-bonded) circuit (Maquet, Getinge, Sweden), and two types of oxygenator (Paragon, Chalice Medical, United Kingdom and Eurosets, Medolla, Italy). Unfractionated heparin was used for anticoagulation, guided by anti-Xa levels (**supplemental document**, <http://links.lww.com/PCC/B785>).

Post-Cannulation Monitoring on ECMO

The patient is monitored continuously, recording saturations from the foot distal to the cannula to observe pulsatility, in addition to assessing perfusion and pulses. The leg is elevated to reduce venous congestion (**Fig. 2B**), and the surgical sites are monitored for bleeding and evidence of infection.

Decannulation and Post-ECMO Care

On discontinuation of ECMO, both cannulae are removed with the intention of repairing both vessels. However in practice, it is usually not possible to repair the FV due to the fragility of tissue in which case this is surgically ligated. Post decannulation, ultrasound of all vessels is performed, and the patient is treated with low molecular weight heparin (LMWH) if there is evidence

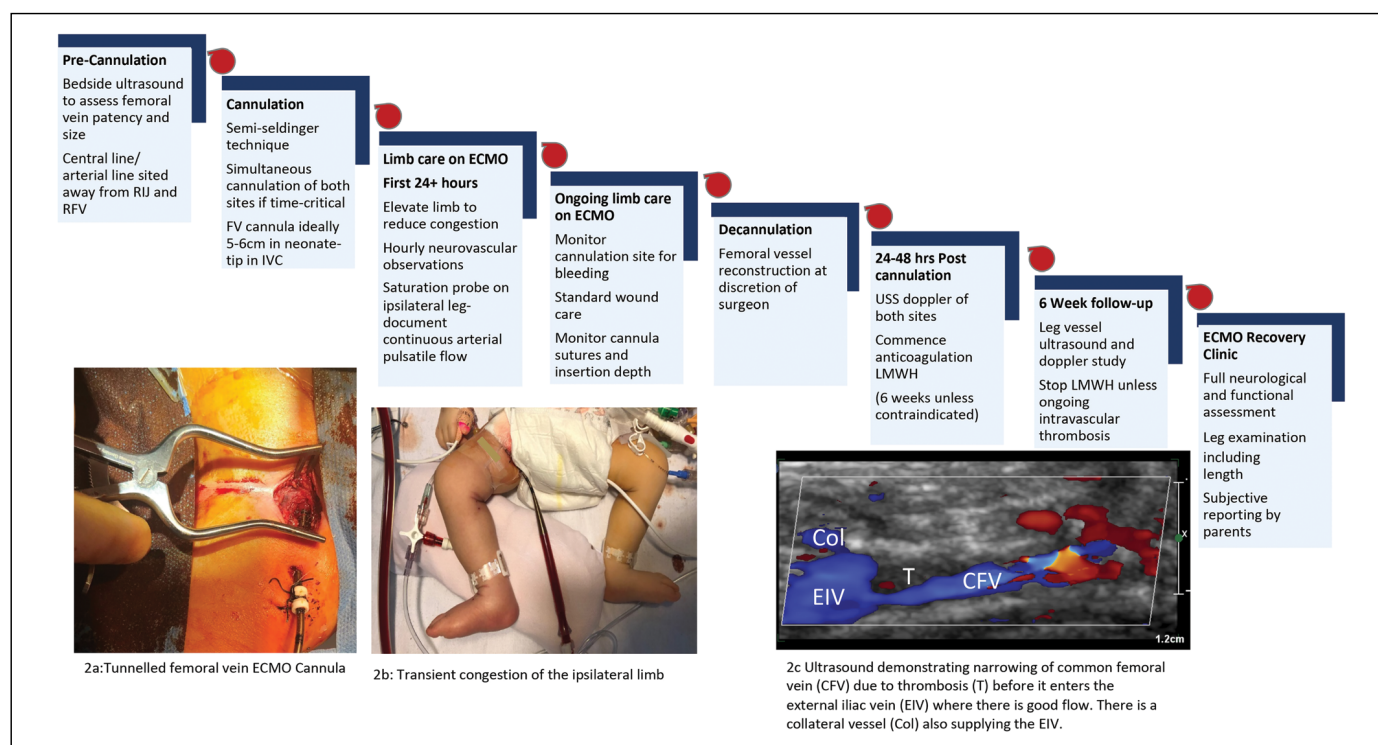


Figure 2. Protocol for femoral cannulation and post-procedure care. **A**, Tunnelled femoral vein extracorporeal membrane oxygenation (ECMO) cannula. **B**, Transient congestion observed post cannulation. **C**, Ultrasound demonstrating narrowing of common femoral vein (CFV) due to thrombosis (T) before it enters the external iliac vein (EIV) where there is good flow. There is a collateral vessel (Col) also supplying the EIV. IVC = inferior vena cava, LMWH = low molecular weight heparin, RFV = right femoral vein, RIJ = right internal jugular vein.

of venous thrombosis (**Fig. 2C**). Ultrasound is repeated at 6–8 weeks post decannulation, and if there is no extension of thrombosis, then LMWH is stopped.

Follow-up is undertaken for all patients until they are 5 years old in the ECMO Recovery Service. This includes assessment and treatment of the child and family by a multidisciplinary team including pediatric specialist occupational therapy consultant, pediatric neurology consultant, ECMO nurse specialist, and ECMO physician, with access to other services as required, including physiotherapy and psychology. Assessment protocol includes developmental and neurologic assessment (Hammersmith Infant Neurologic Examination [10], Bayley Scales of Infant and Toddler Development [11], Vineland Adaptive Behavior Scale [12]), physical examination of the child's surgical sites, and neurovascular examination of the child's leg and length.

Data Collection and Consent

Data regarding patient demographics, clinical status, treatment, radiology, and outcome were collected prospectively for consecutive patients assessed for V_j - V_f from January 1, 2015, to August 31, 2019 with follow-up reported until December 8, 2020. They were analyzed

and reviewed retrospectively with approval from the Evelina London's Service and Evaluation team that consent was not required (Institutional Review Board, Project number 11834). Written, informed consent was obtained for the use of photographs. Calculations and scales presented include oxygenation index (13), Vasoactive Inotrope Score (VIS) (14), and Pediatric Cerebral Performance Category scale (15).

RESULTS

Patients

From January 2015 to August 2019, 12 patients weighing less than 10 kg were assessed for V_j - V_f with all of the survivors followed up for a minimum of 1 year. Eleven underwent V_j - V_p one patient weighing 2.7 kg underwent V-A ECMO due to surgical concerns that the FV was too small to cannulate. This decision was based on the weight of the child rather than appearance on ultrasound. Median weight was 3.6 kg (interquartile range [IQR], 2.8–6.1 kg), and median corrected gestational age 13 days (IQR, 2–175 d) (**Table 1**). The smallest patient weighed 2.1 kg. Primary diagnoses included the following: viral pneumonia ($n = 6$), meconium

TABLE 1.
Summary Table of 11 Patients Less Than 10 kg Receiving Veno-Venous Extracorporeal Membrane Oxygenation

Category	Prevalence/ Median (IQR)
Patient characteristics	
Weight, kg	3.6 (2.8–6.1)
Corrected gestational age, d	13 (2–175)
Comorbidity present	7/11
PCPC	2 (1–3)
Pre-ECMO variables	
Oxygenation index	40 (32–46)
VIS	50 (7–82)
pH	7.1 (7.1–7.3)
Pco ₂ , kPa	8.6 (6.9–10.3)
Cardiac arrest	1/11
ECMO variables	
ECMO flow 24 hr, mL/kg	126 (120–138)
Arterial O ₂ saturation 24 hr, %	94 (91–98)
VIS 24 hr	5 (0–26)
Neurologic injury	2/11
ECMO duration, d	8 (5–12)
Patient outcome	
Persistent neurologic injury	1/11
Survival to home	9/11
Survival to 1 yr	8/11
PCPC of survivors 1 yr	1 (1–2)
Iliofemoral vein patency	
Common femoral	0/9
External iliac	5/9
Limb injury	0/11

ECMO = extracorporeal membrane oxygenation, IQR = inter-quartile range, PCPC = Pediatric Cerebral Performance Category scale (ranked 1–6 where 1 normal, 2 mild disability), VIS = Vaso-active Inotrope Score.

aspiration syndrome ($n = 2$), bacterial pneumonia, alveolar capillary dysplasia, and group B strep sepsis (GBS). Seven patients had comorbidities, five of whom were born prematurely with one still requiring respiratory support.

ECMO Delivery

The right CFV was cannulated successfully in all patients apart from two: one with a previous patent ductus arteriosus ligation via this site and another with a femoral central line in situ. The left CFV was accessed successfully in both of these patients. V_j - V_f was technically successfully in all patients. Adequate flows were achieved with median 126 mL/kg/min (IQR, 120–138 mL/kg/min) (Table 1). Flows were stable with patients waking, allowing neuromuscular blockade to be discontinued in all, and five patients were placed prone without issue. Nine survived to home discharge. Two were palliated due to their underlying disease. One had meningitis due to GBS. Imaging performed once the child was stabilized on ECMO revealed extensive cerebral injury. The other had alveolar capillary dysplasia confirmed on lung biopsy while on ECMO with no option of lung transplant. All children had neurologic imaging while on ECMO (11/11 cranial ultrasound, 7/11 CT head). The majority had normal imaging. Abnormalities were detected in the child with GBS, and another who had venous sinus thrombosis which resolved while being anticoagulated.

One child needed the access cannula repositioned on day 4 of ECMO as the distal end of the RIJ cannula had migrated into the IVC leading to increased recirculation. The patient was extremely edematous at the time of ECMO cannulation when the RIJ access cannula was fixed. By day 4, the cumulative fluid balance was –100 mL/kg, and this led to the cannula migrating inferiorly. The cannula was repositioned surgically with improvement in the patient's saturations.

Limb Follow-Up

Leg swelling due to venous congestion was apparent immediately after cannulation (Fig. 2B), this completely resolved in all cases, and there was never evidence of ischemia. Post decannulation, ultrasound demonstrated complete occlusion of the CFV in all patients, regardless of attempts at vessel repair. There was thrombosis or occlusion identified in the external iliac vein (EIV) in five patients (Table 2). Eight of nine patients subsequently

TABLE 2.
Cannulation, Limb, and Neurologic Follow-Up

Weight (kg)	Ultrasound post decannulation on PICU						Recovery Clinic	
	ECMO Cannulation			Outpatient Follow-Up Ultrasound			Limb Status/ Length of Follow-Up (mo)	Pediatric Cerebral Performance Category Scale at 1 yr
	RIJ Access/ Femoral Return	RIJ	Inferior Vena Cava/Iliac / Femoral Findings	Weeks Post ECMO	RIJ at Follow-Up	Return Findings at Follow-Up		
2.1	12F/8F	Thrombosis	RCFV occlusion, REIV thrombosis	8	Patent	REIV/RCFV occlusion. Collaterals	Normal/22	1
2.4	12F/8F	Patent	RCFV occlusion, REIV thrombosis	4	Patent	REIV/RCFV occlusion. Collaterals	Normal/24	1
2.9	14F/8F	Thrombosis	RCFV occlusion, REIV thrombosis	6	Patent	RCFV occlusion. Collaterals	Normal/15	1
3.6	12F/8F	Thrombosis	RCFV occlusion	7	Occlusion	RCFV occlusion. Collaterals	Stitch abscess healed Normal/12	1
5.5	12F/8F	Occluded	RCFV occlusion	NA ^a	NA ^a	NA ^a	Normal/6 Died 9 mo	6
5.7	14F/10F	Thrombosis	RCFV occlusion, REIV thrombosis, LEIV thrombosis	7	Patent	REIV/RCFV occlusion. Collaterals Left normal.	Normal/18	3
6.5	14F/10F	Occluded	REIV and RCFV occlusions	9	Occlusion	REIV and RCFV occlusion. Collaterals	Normal/18	3
7.0	14F/10F ^b	Patent	LCFV occlusion, RCFV thrombosis	8	Patent	LCFV occlusion, RCFV thrombosis Collaterals	Normal/13	2
8.0	17F/14F	Thrombosis	RCFV occlusion	6	Patent	RCFV occlusion Collaterals	Normal/14	1

ECMO = extracorporeal membrane oxygenation, LCFV = left common femoral vein, LEIV = left external iliac vein, NA = not applicable, RCFV = right common femoral vein, REIV = right external iliac vein, RIJ = right internal jugular vein.

^aPatient did not have outpatient ultrasound follow-up.

^bReturn cannula via left femoral vein as cannula would not advance along right.

received a course of LMWH and a follow-up ultrasound (median of 7 wk post decannulation). Lower limb vessels remained occluded despite LMWH, but all patients had developed collateral vessels with excellent flow into the IVC, so anticoagulation was stopped. There was no evidence of IVC thrombosis in any patient. At least 1 year post ECMO, there were no reported signs or symptoms in the ipsilateral leg nor difference in diameter

or leg length. One child had a sterile stitch abscess of the inguinal region which healed once the stitch was removed. One child died 9 months post ECMO due to chronic lung disease after recurrent chest infections and repeated admissions to PICU.

At follow-up to at least 1 year post ECMO, neurologic assessment was normal in five of eight of the survivors. Three children had developmental delay, but

they all had significant comorbidities that preexisted ECMO and may account for delayed milestones.

DISCUSSION

This study demonstrates that V_j - V_f can be safely established in neonates and infants as small as 2.1 kg, using the CFV for the return cannula. Adequate ECMO flows (median 126 mL/kg/min) and patient saturations were achieved in a heterogeneous population of nonambulatory children of whom seven had comorbidities including chronic lung disease, pulmonary hypertension, and immunodeficiency. Seven patients had a significant inotrope requirement ($VIS \geq 30$) with none requiring conversion to V-A ECMO; they received higher levels of support than most patients in case series reporting the use of ECMO in inotrope-dependent neonatal and pediatric respiratory patients (16, 17). This cannulation method achieved flows that were stable, with minimal recirculation on ECMO runs extending to 28 days and included some patients being positioned prone.

The main complication observed in this series was that the CFV was sacrificed with occlusion of the EIV in four of the surviving patients. Examination of the leg was normal in all cases, but no flow was observed on Doppler ultrasound of the CFV post decannulation, regardless of vessel repair. Repeat ultrasound, at a median interval of 7 weeks, after a course of LMWH demonstrated good collateralization and no intravascular thromboses in the iliac vessels and IVC. Follow-up to at least 1 year post decannulation demonstrated no medium-term effects of the CFV being obstructed/ligated with equal leg growth and no muscular nor neurovascular deficit. LMWH was administered to all patients post decannulation to reduce the risk of thrombogenesis in adjacent vessels, but there is no control group to offer comparison or assess the impact of this strategy. There is limited literature on the prevalence of cannula-associated deep vein thrombosis (DVT) post ECMO cannulation with a paucity of data in the infant population. In adults, the prevalence of DVT post ECMO is as high as 85% (18).

The first case series of neonatal V_j - V_f in 1985 reported issues with leg congestion, recirculation, and longer operative time to place a second venous cannula than V-A ECMO (9), and so the authors concluded that V-A ECMO was preferred to V-V.

However, this original case series showed that V_j - V_f was effective, with high rates of survival, and the only significant complication—a retroperitoneal bleed—was reported in a neonate weighing 1 kg. Our case series demonstrates that although V_j - V_f in smaller babies does cause initial venous congestion with CFV occlusion post decannulation, there is no evidence of morbidity in the longer term. To our knowledge, there are two other ECMO centers worldwide using this method in neonates; however, they have not published their case series with regard to complications (19) (W. Butt, personal communication, 2015).

We suggest that occlusion of the CFV is preferable to the increased risk of neurologic injury observed in V-A ECMO patients (1). An additional benefit of V-V ECMO is the option of reducing anticoagulation if bleeding complications arise. In one patient, heparin was stopped for greater than 24 hours after prolonged haemorrhage secondary to a lung biopsy. The risk of running heparin-free would have been higher with V-A ECMO using the carotid artery. The significant increased risk of neurologic injury with V-A ECMO should not be trivialized in order to justify using V-A ECMO to support neonates with respiratory failure (4, 20). Over decades, V-V ECMO has been shown to reduce complications compared with V-A by using VVDL, but now confidence has been reduced in this cannulation strategy, an alternative technique should ideally be adopted.

The introduction of VVDL in the 1990s meant that focus has been on VVDL in the neonatal population with use of V-V ECMO rising to as high as 86% in the United Kingdom in the 2000s (8). The largest case series of neonates treated with V-V ECMO via VVDL reported a risk of cardiac perforation of 7% with perforation identified 12–144 hours into the ECMO run rather than at the time of cannulation (5). This publication highlighted a problem faced by other centers that also included concerns with the difficulty in placing the VVDL correctly, so that the distal end was in the IVC. These concerns are echoed internationally and reflected in the percentage of neonates with respiratory failure being supported by V-V ECMO declining to 16% of cases worldwide—the lowest in over 20 years.

VVDL have been used safely in other single-center case series using percutaneous insertion and extensive use of ECHO to place and monitor the cannulae (21)

or a combination of ultrasound and fluoroscopy (20), but these report fewer neonatal cannulations than the U.K. study. Worldwide, the rate of cardiac perforation with VVDL in pediatrics was estimated to be 4% with highest risk in the neonatal population (6).

Comparison of V_j - V_f and VVDL has been limited to outside of the neonatal population with lower mechanical and cardiovascular complication rates demonstrated in multisite cannulation in children (22). There are no data to suggest whether VVDL or V_j - V_f is better in the neonatal population. Since many centers are no longer offering VVDL to neonates, V_j - V_f should be explored.

LIMITATIONS

Although the data from our case series are comprehensive, it include only 11 patients, and further longitudinal evaluation of V_j - V_f in this population is required to determine the prevalence of complications. Children will all be followed up until at least 5 years old.

The cannulation method only included surgical insertion, and it is not known whether the femoral occlusion rate could be reduced with percutaneous cannulation. This center aims to place cannulae percutaneously in the future using intensivists/interventional cardiologists.

CONCLUSIONS

V_j - V_f was performed safely in patients under 10 kg with the smallest weighing 2.1 kg. Although there was occlusion of the CFV in patients post decannulation, subsequent follow-up demonstrated no clinical implications. We challenge the belief that V_j - V_f cannot be performed in nonambulatory patients and suggest that this strategy is preferred over V-A ECMO in infants requiring ECMO for respiratory failure. Further evaluation of this technique is required with regard to comparison with VVDL.

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